

NEW RIVER VALLEY, VA, USA

WMO: 724116

Lat: 37.133N

Lon: 80.683W

Elev: 2105

StdP: 13.61

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13868

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 12.2	(c) 17.6	(d) -1.8	(e) 5.4	(f) 19.8	(g) 2.7	(h) 6.8	(i) 21.8	(j) 26.8	(k) 33.4	(l) 24.6	(m) 33.6	(n) 7.8	(o) 300	(p) 0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.9	(c) 90.4	(d) 70.7	(e) 88.2	(f) 70.5	(g) 85.7	(h) 70.0	(i) 74.9	(j) 84.5	(k) 73.6	(l) 82.3	(m) 72.5	(n) 80.9	(o) 6.4	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.3	129.2	78.6	71.5	125.8	77.9	70.1	120.0	76.6	39.7	84.7	38.7	82.1	37.6	81.0	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.7	(b) 18.7	(c) 16.4	(e) 6.2	(f) 93.5	(g) 5.8	(h) 2.9	(i) 2.0	(j) 95.6	(k) -1.4	(l) 97.2	(m) -4.6	(n) 98.9	(o) -8.8	(p) 100.9		
(a) 21.7	(b) 18.7	(c) 16.4	(e) 4.9	(f) 76.6	(g) 5.6	(h) 1.4	(i) 0.8	(j) 77.6	(k) -2.5	(l) 78.5	(m) -5.6	(n) 79.3	(o) -9.7	(p) 80.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.6	35.4	38.4	45.8	55.4	63.9	71.2	74.4	73.2	67.0	56.5	45.8	38.6
	DBStd	15.86	10.27	9.82	9.69	8.39	7.20	5.18	4.27	4.26	6.14	7.78	8.63	8.92
	HDD50	1625	464	340	199	42	3	0	0	0	30	180	367	
	HDD65	4406	918	744	598	300	110	12	1	2	51	278	575	817
	CDD50	3655	11	16	67	204	434	637	756	719	509	232	55	15
	CDD65	963	0	0	1	12	76	199	293	256	110	15	1	0
	CDH74	8095	0	1	18	198	710	1664	2430	2030	881	155	8	0
	CDH80	2581	0	0	1	32	161	532	884	695	250	26	0	0
(14) Wind	WSAvg	5.5	7.6	7.5	7.5	7.1	5.3	4.3	3.7	3.1	3.5	4.7	5.8	6.3
(15) Precipitation	PrecAvg	39.5	3.2	2.5	3.7	3.1	4.3	3.9	4.2	3.3	3.1	2.5	2.6	3.1
	PrecMax	53.5	6.7	6.3	7.2	5.1	9.0	8.5	7.1	6.1	8.9	6.7	4.6	6.3
	PrecMin	30.5	1.3	0.8	1.1	0.9	1.9	1.1	1.7	1.5	0.3	0.1	0.4	1.3
	PrecStd	5.6	1.4	1.2	1.5	1.1	2.0	1.9	1.4	1.2	1.9	1.7	1.2	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.6	69.9	76.8	83.9	88.1	91.3	93.3	91.5	90.1	82.7	74.5	66.4
		MCWB	57.0	57.6	58.9	61.7	67.5	70.8	71.3	70.6	66.5	65.5	59.3	57.9
	2%	DB	60.9	63.7	72.0	79.4	83.9	88.5	90.4	89.9	86.0	78.8	69.6	61.5
		MCWB	54.4	52.9	56.5	60.9	67.2	70.9	71.5	71.1	68.1	63.2	56.2	55.6
	5%	DB	55.4	58.9	66.1	75.2	81.4	85.8	88.0	87.1	82.3	73.9	64.1	57.1
		MCWB	49.9	51.1	53.7	59.2	65.7	69.9	71.2	71.1	67.4	61.3	55.4	52.3
	10%	DB	51.8	54.4	62.6	71.9	78.6	82.4	84.5	83.9	79.4	70.3	61.0	53.8
		MCWB	46.7	47.6	52.1	57.7	65.2	68.5	70.6	70.0	66.4	60.2	53.1	49.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.9	60.5	62.2	66.5	71.5	75.5	76.5	75.8	73.8	70.8	63.4	61.5
		MCDB	63.5	66.8	71.8	77.9	81.3	87.1	86.9	85.8	81.3	77.0	67.7	65.0
	2%	WB	56.0	56.6	59.3	64.1	69.8	73.4	75.1	74.4	71.8	67.5	60.3	57.4
		MCDB	60.0	61.4	67.0	74.6	79.1	82.6	84.6	83.6	79.3	73.5	65.3	59.9
	5%	WB	51.7	52.7	56.8	62.0	68.2	71.9	74.0	73.1	70.2	64.5	57.8	54.0
		MCDB	55.0	57.0	64.0	71.5	77.3	81.3	82.8	81.7	77.6	70.3	62.9	56.6
	10%	WB	46.5	48.2	54.0	59.6	66.7	70.5	72.8	71.8	68.7	62.1	54.6	49.4
		MCDB	49.5	53.5	60.1	67.7	75.1	79.6	81.1	80.2	75.7	68.4	59.2	52.8
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	18.3	20.5	22.4	22.0	20.9	19.9	20.3	20.9	21.7	20.5	17.2
		MCDBR	23.0	26.1	27.7	28.9	26.6	24.9	24.1	24.7	25.8	27.4	26.9	23.5
		MCWBR	16.6	17.5	15.5	14.2	12.8	9.9	9.1	9.6	10.6	13.9	16.9	17.0
	5% WB	MCDBR	20.2	21.9	22.7	24.0	22.5	21.5	20.0	20.6	20.3	21.0	20.9	19.8
		MCWBR	16.8	17.4	14.8	13.5	11.6	9.6	8.7	8.8	9.3	12.4	15.8	16.8
(40) Clear-Sky Solar Irradiance	taub		0.295	0.307	0.332	0.373	0.414	0.441	0.473	0.479	0.377	0.344	0.309	0.293
	taud		2.502	2.459	2.435	2.341	2.231	2.187	2.129	2.087	2.373	2.459	2.522	2.538
	Ebn at Noon		286	295	296	288	276	268	258	253	277	278	278	279
	Edh at Noon		25	30	34	39	44	46	49	50	36	30	25	23
(44) All-Sky Solar Radiation	RadAvg		705	922	1260	1603	1767	1924	1802	1656	1395	1102	822	612
	RadStd		60	112	82	129	147	137	113	85	135	135	80	58

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+250	+148

Nomenclature: See separate page